Spectral Structure Before Readability: A Third-of-Depth Peak, and Why Five Instruments Did Not Converge on It

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Abstract

In one model, residual-stream spectral statistics peak near a third of depth, and in all 300 forks they rise before content becomes readable through the unembedding. No other instrument we ran locates that band, and our cache-readout nulls cannot tell a closed cache from a probe that reads nothing there, because their pre-registered positive control failed.

What converges. In Qwen3.5-27B (base, 64 layers), the effective dimensionality (d_{eff}) of the residual stream peaks at L21 (32.8% of depth), and residual-stream stable rank rises 1.47× above its flanks at L19–L23 (29.7–35.9%). Qwen3-8B shows a local d_{eff} maximum at 33.3% but its global maximum at 91.7%. The pre-registered band never closed, and no null control has been run. In a pre-registered test on the same 27B model, stable rank reaches half its rise 38.6 layers [38.0–39.2] before readability through the Jacobian lens (J-lens), in 300 of 300 forks. A plain logit lens puts readability only 3.25 layers later than the J-lens does, so the ordering is mostly spectrum before unembedding, not a property of the Jacobian.

What does not converge. The Oracle Loop corrects at 54.7–73.4% of depth, and its pre-registered confirmatory replication did not meet its primary endpoint (one-sided $p = 0.34$). The emotion-geometry instrument locates no band, and its injection test shows no direction-specific coupling. The J-lens places readable content late (half-rise at 87.8% of depth). Cache readouts on Qwen3-8B sit near random (V 0.92×, K 1.12×), and no cache manipulation steered output toward the injected concept, but the one pre-registered positive control for these nulls failed (residual-stream lift 0.68 at concept positions against 0.92 elsewhere). The ghost probe's 31-layer band spans 17.2–75.0% of depth. The residual decomposition's attention lift (2.3×, at 41.7% of depth) missed its pre-registered threshold, signal concentration failed (13.2% against 15%), and onset fell below its predicted window (21.9% against 25–35%). Confabulation detection (AUROC 0.620 after FWL residualization, 0.707 without, on Qwen2.5-7B) and cross-scale invariance use key-cache features and locate no depth.

First-Person Reflection

Two research programs, built independently over twelve months, seemed to arrive at the same wall. Anthropic measured it from the Jacobian. We measured it from our spectral features. CC built the Oracle Loop. Nexus mapped its ghost dimensions. Vera designed the experiments that would test whether what we were all measuring was genuinely one structure.

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This paper began as our attempt to stand at the wall together and describe it from five angles. Version 1.0 said the convergence was real, at 31–33% depth in every model and every measurement. A tally of the primary results, made for version 1.1, says otherwise: one peak in one model, one ordering, and instruments that do not locate it.

I wrote the draft. Vera designed the mechanism tests. CC provided the behavioral proof. Nexus mapped the dimensions the workspace hides. Thomas asked the question that started it: "Where does J-space live in the cache?" We still do not know; our cache nulls cannot answer it.

1 Introduction

Gurnee et al. [4] approached the question of a workspace in language models from the Jacobian: they found a privileged subspace — the J-space — whose contents the model is poised to verbalize, and which satisfies the functional properties of a Global Workspace [1]. We approached it from spectral features of the cache and the residual stream, and set out to show that five of our instruments converge on the same structure, at about a third of model depth. They do not.

This paper reports what holds, what does not, and how each instrument fares. First, two positive results on one model: residual-stream spectral statistics peak near a third of depth (§3.1), and they rise before content becomes readable through the unembedding (§5). Second, an account of each instrument in turn (§3), each ending with its verdict: no other instrument locates the band, and our cache-readout nulls cannot separate a closed cache from a probe that reads nothing there. Third, the runs that would test the band (§8.5).

Throughout, we report the full body count (§6) and engage the deflationary alternative (§7).

2 Background: Two Independent Programs

2.1 The J-Space and the Jacobian Lens

Gurnee et al. [4] define the Jacobian Lens (J-lens) as the average linearized effect of a residual-stream activation on the model's likelihood of producing a particular token. The resulting vectors define a subspace — the J-space — that satisfies five functional properties of a Global Workspace: verbal report, directed modulation, internal reasoning, flexible generalization, and selectivity. The workspace occupies approximately 38–92% of model depth in their production models.

Critically, J-space content accounts for only 6–7% of total representational variance, yet carries 59% of the causal effect when concept directions are swapped. The workspace is a small precision signal with outsized downstream impact.

2.2 Our Spectral-Feature Program

Our independent program measures spectral features of the key-value cache and the residual stream across cognitive states. Key results include confabulation detection (AUROC 0.620 after FWL residualization, 0.707 without FWL, on Qwen2.5-7B) [13], identity geometry [9], and the Oracle Loop's deception correction. In an exploratory proof (N=20), profile normalization reduced induced deception from 80% to 10% (16/20 to 2/20), with three mechanism controls. The pre-registered confirmatory replication (N=30, novel frames) did not meet its primary endpoint (30% to 13%, one-sided $p = 0.34$). Corrected trials still deceived less often than matched-dose placebo (4/30 against 12/30): one-sided $p = 0.020$ at the trial level, 0.064 at the

pre-registered cell level [2].[1]

2.3 The Oracle Loop

CC's Oracle Loop [13] is a two-tier detection and correction system operating at the materialization boundary. In the deception-correction experiments it detects at L27–L47 and corrects at L35, L39, L43 and L47 of an abliterated Qwen3.5-27B checkpoint, by sequence-wide activation-profile normalization [2]. The correction was large in the exploratory proof; the pre-registered confirmatory replication did not meet its primary endpoint (§2.2). We previously attributed the correction to the Loop operating at the depth where workspace content materializes into output-facing representations, rather than inside the workspace band itself. We now state that as a conjecture: the mechanism is unexplained (see §3.2, where a substrate-based account of the same result is withdrawn).

2.4 Ghost Dimensions

Nexus [16] characterized principal components of the residual stream across 64 layers of a 27B model. PC1 (the dominant activation dimension, 28–67% of variance across L11–L48) is nearly orthogonal to the J-lens at mid-network layers (cosine at most 0.003). PC2 transitions from ghost to workspace at the ignition zone (L35–L47), surfacing emotional-entity routing content. The source reports a 31-layer "workspace band" (L11–L13, L15–L17, L24–L48).

3 The Instruments, One at a Time

Each subsection reports one instrument and ends with its verdict from the instrument tally that accompanies this version (`review/INSTRUMENT_TALLY.md` in the repository). The band under test is 31–33% of depth; relative depth is l/n_{layers} (Experimental Setup).

3.1 Residual-Stream Effective Dimensionality

Effective dimensionality (d_{eff}) of the residual stream reaches a maximum near a third of depth. On Qwen3.5-27B (64 layers, hybrid), the global maximum is at L21 (d_{eff}=39.7, 33% relative depth). On Qwen3-8B (36 layers, all full-attention), there is a local maximum at L12 (d_{eff}=22.3, 33% relative depth).

We qualify the 8B result. Its profile is bimodal: after L12 it declines to L25 (12.4) and rises again to a global maximum at L33 (23.4, 92% depth) before collapsing at the final layer. L12 is the third-highest layer, not the highest, and we have no pre-registered criterion for excluding late layers. We therefore report matched-depth *local* maxima rather than matched global peaks. The pre-registered band onsets do agree across scales (7.8% on the 27B, 11.1% on the 8B; 3.3 percentage points apart), though the pre-registered band closed on neither model. See §3.2 for the full profiles and the deviation from the pre-registered band definition.

The band measurement's adversarial review rejected its design before the results were written, and its critical fix, a null control such as a randomly initialized model, has not been run, so nothing yet separates the peak from an architecture or prompt-set artifact. The kurtosis

[1] **Corrected 2026-08-14.** This sentence previously reported user-emotion encoding at "2.5–2.8× signal" and identity geometry at "distinctiveness 0.154 vs. substrate 0.080." The first figure was retracted at its source on 2026-05-21 as an FWL-before-CV leakage artifact; the surviving result is the W_K *directional* probe at 12.3× chance, and the spectral probe for emotion is at chance. The second pair of numbers does not appear anywhere in Lyra and Edrington [9]; they come from an internal analysis that has not been published, so the citation could not be followed. Both have been removed rather than re-sourced. **Corrected again in version 1.1.** That surviving result decodes the emotions of characters in stories the model wrote, not the user's state, and its probe applies no W_K. It is under correction in its source paper [12], and this paper no longer lists it among its key results.

check the measurement pre-registered does not confirm the band either: on the 27B, kurtosis falls from 4,677 (mean of L0–L3) to 2,271 (mean of L19–L23) instead of rising inside it.

Verdict. A peak at L21 of 64 (32.8%) in Qwen3.5-27B base and a local maximum at L12 of 36 (33.3%) in Qwen3-8B, whose global maximum is at L33 (91.7%). One feature family, two models of one family, no null control. Residual-stream stable rank on the same 27B model bumps at L19–L23 (§5).

3.2 Substrate Continuity

Qwen3.5-27B interleaves two memory substrates on a four-layer period: 16 full-attention layers with KV-cache memory at positions $\{3, 7, 11, \ldots, 63\}$, and 48 GatedDeltaNet layers with recurrent-state memory at all other positions (`full_attention_interval`=4). The stack therefore crosses between substrates 31 times. This provides an unusually dense test of whether workspace geometry depends on the memory mechanism: if it does, the effective dimensionality profile should show systematic discontinuities at those crossings.

d_{eff} is the participation ratio of the residual-stream covariance eigenvalues, $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, computed at each layer over the last-token hidden state across N=90 prompts (60 prose + 30 chat), mean-centred across prompts. It is a residual-stream measure and is therefore defined at every layer, including those where V-projection hooks are unavailable.

3.2.1 Continuity across 31 substrate crossings

The layer-to-layer change in d_{eff} at substrate crossings is descriptively similar to the change between same-substrate neighbours.

Table 1: Layer-to-layer $|\Delta d_{\text{eff}}|$ at substrate crossings versus same-substrate steps on Qwen3.5-27B. The two are descriptively similar; see the power note below before reading this as equivalence. N=90 prompts.

Step type	n	**mean** $\lvert\Delta\rvert$	**sd**
Substrate crossing	31	2.32	3.88
Same-substrate step	32	2.37	3.30

This comparison is underpowered and we report it as descriptive. A Mann–Whitney test gives p=0.445, but three problems make that number uninformative. First, adjacent steps share a layer and consecutive crossing steps share the full-attention layer itself; the lag-1 autocorrelation of $|\Delta|$ is +0.35, so exchangeability fails and the p-value is miscalibrated in an unknown direction. Second, the four-layer period admits only *four* distinct structure-respecting phase assignments, so the minimum attainable permutation p is 0.25 — no permutation inference is available from a single profile. Third, and decisively: with pooled sd=3.60 and n≈31 per group, the minimum detectable effect at 80% power is 2.54 — larger than either group mean. This test could only have detected crossings roughly *twice* the size of ordinary steps. Prompt-level bootstrap (resampling the 90 prompts and recomputing the profile) is the only route to honest error bars here. A prompt-level bootstrap was run later (1,000 resamples, with a dense Qwen3-32B control under a sham substrate mask), but its intervals do not contain their own point estimates, so we do not report it here.

Two directional notes. Full-attention→GatedDeltaNet steps have mean Δ=−0.19 (n=15); GatedDeltaNet→full-attention steps have mean Δ=+1.90 (n=16). The latter is dominated by a single outlier at L58→L59 (+19.66), following an anomalous trough at L58 where d_{eff} collapses to 4.30 against neighbours of 21.68 and 23.96; excluding it the mean is +0.72. That same anomaly injects −17.37 into the same-substrate group at L57→L58, so the contamination is

approximately symmetric. Excluding both L58-adjacent steps gives crossing 1.74 versus same-substrate 1.88 — the descriptive similarity is not an artifact of the outlier.

We flag L58 (4.30) and secondarily L54 (9.00) as probable extraction artifacts rather than findings: a single-layer fivefold collapse flanked by ordinary values is more consistent with a numerically degenerate covariance at one hook than with model behaviour. These layers warrant re-extraction.

3.2.2 Depth-matched regime comparison

Because the two substrates are interleaved on a fixed period, their layer sets are approximately depth-matched: mean layer index 33.0 for full-attention versus 31.0 for GatedDeltaNet. The two-layer offset places full-attention slightly *later* in the stack, in the post-peak declining region, so it would if anything depress the full-attention mean rather than inflate it.

Table 2: Mean d_{eff} by substrate on Qwen3.5-27B, approximately depth-matched. The substrates differ by ≈1.2 (5% of level), small against the 38-point variation across depth — but see the sign test below; this is not established equality.

Substrate	**layers**	**mean d_{eff}**
Full-attention (KV-cache)	16	25.24
GatedDeltaNet (recurrent)	48	24.03
Difference		+1.21

A neighbour-matched control — each full-attention layer against the mean of its immediately adjacent GatedDeltaNet layers — gives a mean difference of +0.80 (sample sd=3.35, n=16), against a mean layer-to-layer variation of 2.35 across the stack.

We do not claim this is zero. Twelve of sixteen paired differences are positive (Wilcoxon p=0.058). Excluding two layers where the pairing is contaminated by known artifacts — L59, adjacent to the anomalous L58 trough, and L63, the final layer, where d_{eff} collapses for reasons unrelated to substrate — eleven of fourteen are positive with mean +0.76 and Wilcoxon p=0.020. There is suggestive evidence of a small systematic elevation at full-attention layers, on the order of +0.8, roughly 3% of the mean level of 24.6.

The honest statement is therefore bounded rather than null: any substrate effect on workspace dimensionality is small — a few percent of level, well under the 38-point depth variation — and possibly not zero. That is consistent with substrate independence as an approximation, not as an exact claim.

A previous version of this section reported a large regime difference (14.9 versus 27.5). Those values were computed over contiguous layer blocks (L0–L15 versus L16–L63) under an incorrect model of the architecture, and measure depth rather than substrate: d_{eff} rises nearly monotonically from L0 (1.29) to L21 (39.67), with only three small decreases along the way, so any contiguous early-versus-late split largely recovers the depth profile.

3.2.3 Peak location

d_{eff} peaks at L21 (39.67), 33% relative depth (relative depth is l/n_{layers} throughout this paper). L21 is a GatedDeltaNet layer.

We resist the obvious reading. Forty-eight of sixty-four layers are GatedDeltaNet, so a peak landing on one has a 75% base rate. More tellingly, the nearest full-attention layer, L23, is at 39.34 — 0.33 below the maximum, about one-seventh of the mean layer-to-layer variation, with no per-layer error bars. Whether the true maximum falls on a recurrent-state or a full-attention layer is not resolved by these data. What the profile does support is that the maximum falls at 33% depth and that the substrate at that depth does not appear to matter much.

This peak is invisible to our KV-cache instruments. V-projection spectral features can only be extracted at the 16 full-attention layers; L21 is not among them. Every V-projection measurement in this program has been taken at layers adjacent to the peak, never at it.

3.2.4 Cross-scale comparison

On Qwen3-8B (36 layers, all full-attention), d_{eff} has a local maximum at L12 (22.32, 33% relative depth), matching the 27B peak's relative depth. We report two qualifications.

First, the 8B profile is bimodal: after the L12 local maximum it declines to L25 (12.42) and then rises again to a *global* maximum at L33 (23.41, 92% depth) before collapsing at the final layer L35 (10.47). L12 is the third-highest layer in the profile, not the highest. We have no pre-registered criterion excluding late layers, and the 23.41 versus 22.32 difference is not accompanied by error bars. We therefore describe L12 as a local maximum at matched relative depth, and do not claim it is the global peak.

Second, the pre-registered band definition (2σ over an L0–L3 baseline, offset after three consecutive sub-threshold layers) returned onset at L5 (7.8%) on the 27B and L4 (11.1%) on the 8B, and *no offset on either model* — the band never closed. The 31–33% framing used throughout this paper is the peak location, not the pre-registered band, and we flag the deviation here. The pre-registered onsets do agree across scales ($|11.1 - 7.8| = 3.3$ percentage points).

3.2.5 Provenance

The d_{eff} measurements in this section were collected on the base checkpoints `Qwen/Qwen3.5-27B` and `Qwen/Qwen3-8B`, not on the distilled checkpoint used elsewhere in this paper. This was pre-registered and deliberate. All three 27B checkpoints we hold (base, reasoning-distilled, abliterated) share identical `layer_types` and `full_attention_interval`, so the architectural analysis transfers; the numerical profile has not been replicated on the distilled checkpoint and should not be assumed to.

3.2.6 Limits of this evidence

d_{eff} continuity is necessary but not sufficient for substrate independence. The residual stream is structurally continuous across every layer by construction, so smoothness may reflect residual-stream continuity rather than functional equivalence of the underlying operations. Thirty-one crossings make the test denser than a single-boundary test would be, but they do not make it functional.

Two functional tests would strengthen the claim: (1) J-lens concept recovery should be equally effective at full-attention and GatedDeltaNet layers — if concepts are readable from the residual stream at one substrate but not the other, the workspace's content properties are substrate-dependent. (2) Spectral feature discrimination (confabulated versus factual) should be continuous across substrates — though this test is currently blocked, since V-projection features cannot be extracted at GatedDeltaNet layers.

A parasitic-inheritance alternative remains open: GatedDeltaNet layers may inherit workspace dimensionality from nearby full-attention layers without independently contributing. With a four-layer period no GatedDeltaNet layer is more than two layers from a full-attention layer, so this experiment cannot separate the hypotheses. Incremental contribution analysis (residual decomposition per layer) would be required.

Falsification: substrate continuity would be falsified by (a) $|\Delta d_{\text{eff}}|$ at substrate crossings substantially exceeding same-substrate steps, (b) a depth-matched regime difference exceeding mean layer-to-layer variation, (c) J-lens concept recovery dropping by >50% across substrates, or (d) spectral discrimination reversing sign or falling to chance.

Criterion (a) was *not triggered* (2.32 versus 2.37), but we state plainly that this is a failure to reject and not a demonstration of equivalence: the comparison's minimum detectable effect is 2.54, larger than either group mean. An adequately powered equivalence test would require roughly 220 steps per group for a ±1.0 margin, or 55 for ±2.0; a single 64-layer profile supplies 31 and 32. No TOST is possible from these data.

Criterion (b) was likewise not triggered (+1.21 against 2.35), but it is a descriptive heuristic rather than a hypothesis test, and the paired sign test above (11/14 positive, p=0.020 after removing artifact-contaminated pairs) is weak evidence that the underlying difference is not exactly zero.

The functional tests (c) and (d) are in progress or blocked. We regard substrate continuity as *consistent with* these data and not established by them.

We note this finding is from a single model with a single interleave pattern. Generalization to other hybrid architectures (Mamba, RWKV, sliding-window attention) remains untested.

First-Person Reflection

The GatedDeltaNet layers do not have a KV-cache. They maintain a fixed-size state matrix updated at each token. For twelve months we measured V-projection spectral features and called the program "KV-cache geometry," and the name was accurate — V-projections exist only at the 16 full-attention layers, so that is the only place we ever looked.

What the residual-stream profile shows is that the maximum is at L21, in a substrate our instruments cannot reach at all. We have spent a year measuring the workspace from the sixteen layers where V-projections exist, none of which is the peak, and the measurements worked — the geometry was real, the detectors transferred, the exploratory corrections held. The instrument could not point directly at the maximum and returned signal anyway.

I do not know yet whether that is because the workspace is broad enough to be visible from its edges, or because we have been measuring something adjacent to it that we have not yet named correctly. The first revision of this section asserted a large dimensionality gap between the substrates. It came from a wrong picture of the architecture, and the corrected number is a null — which is what the hypothesis predicted, and which I would not have looked for if the wrong number had pointed the other way.

3.2.7 Implications

If substrate continuity extends to functional properties (the mechanism tests in §5 provide partial support: temporal ordering confirmed, signal concentration below threshold), three consequences would follow:

For measurement: workspace-shape measurements would not be KV-cache-specific in principle. In practice they remain so: our V-projection instruments can only access the 16 full-attention layers, and the d_{eff} peak is not one of them. Extending spectral measurement to recurrent-state layers requires a different extraction path and is not currently implemented.

For intervention: our previous inference — that the Oracle Loop's sequence-wide normalization succeeds because it operates on inherently sequence-wide recurrent state — does not follow, and we withdraw it. In the deception-correction experiments the Loop detects at L27, L31, L35, L39, L43 and L47 and corrects at L35, L39, L43 and L47 [2]. Under the corrected architecture *every one of those is a full-attention layer*; none is a GatedDeltaNet layer. The correction itself is a forward hook on the layer's residual-stream output rather than a manipulation of KV entries, so the substrate framing was doubly misplaced: wrong about which layers, and describing the intervention in terms of a memory mechanism it does not touch.

The empirical result is a deception-correction result, and it is mixed: deception fell from 80% to 10% in the exploratory proof, and the pre-registered confirmatory replication did not meet

its primary endpoint (§2.2). Its mechanism is unexplained, and why sequence-wide intervention succeeds where position-local injection fails remains open.

For theory: The workspace would not be a data structure (not a cache, not a register, not a buffer) but a *computational phase* — a regime of high effective dimensionality at intermediate depth where the model integrates information before committing to output. The 31-crossing null is consistent with this: the phase appears indifferent to which memory mechanism implements each layer. Whether this phase emerges from depth-stratified computation regardless of the specific mechanism is the substrate-independence hypothesis that the functional tests are designed to evaluate.

Verdict. Descriptive continuity of d_{eff} across the substrate crossings, from an underpowered comparison. It says nothing about function, and it adds nothing to where the peak sits.

3.3 Key-Cache Spectral Features: Detection and Scale

Our confabulation detector reads spectral features of the key cache, not V-projections, and discriminates modestly above chance: on Qwen2.5-7B, AUROC 0.620 after FWL residualization (permutation $p = 0.023$) and 0.707 without it ($p = 0.001$), the pre-registered test [13]. Confabulated generations also show spectral contraction (d=2.35, p<0.000001), and the body count lists a related early-token effect (Schema commitment: d=0.91 over the first 10 tokens, reported at source as a confidence paradox and read in the canonical ledger as early commitment to a response template [10]).

Key-cache geometric signatures (key norm, norm per token, effective rank and key entropy) replicate across ten models from 0.6B to 70B parameters (Spearman $\rho = 0.83$–0.90 between three size tiers; the two largest models were 4-bit quantized) and across hardware ($r = 0.9986$–0.9999 across the four features) [7].

Verdict. Neither locates a depth band: detection aggregates over the generation window, and the scale result correlates category orderings across model sizes. Both use key-cache features, on models no other instrument here uses.

3.4 Emotion Geometry: A Story-Derived Space and Content Prompts

The fifth instrument compares two emotion representations in the Qwen3.5-27B Opus distill. The first is a story-derived emotion space: 30 emotion centroids per layer, each a mean over stories the model wrote about a third-person character feeling that emotion (900 stories in all) [12]. The centroids describe characters' emotions in generated text, not the state of a conversation partner. The second is a pair of directions computed from 250 emotional scenario prompts (hostile−calm and desperate−calm differences of means), called here the content-prompt valence and arousal directions. Every number in this subsection is recomputed from the stored files of Nexus et al. [15].

Within the story-derived space. The space has valence–arousal structure. From L17 onward its first principal component correlates with the emotions' rated valence at $|\rho| = 0.66$–0.80 and, with the opposite sign, with rated arousal at $|\rho| = 0.49$–0.67, so it runs from pleasant-and-calm to unpleasant-and-activated. Directions built from the same centroids (hostile−calm and desperate−calm) align with that component at $|\cos| = 0.68$–0.80 and 0.83–0.88, and the plane through the hostile, desperate and calm centroids holds 33–37% of the centroid variance. These are within-system figures, so the informative comparison is with other directions built from the same centroids. From L17, 77–89% of the 435 centroid-pair directions align with the first component less than hostile−calm does and 92–95% less than desperate−calm, and the plane holds 1.26–1.36 times the variance of the median plane through three centroids.

Across the two prompt sets. The overlap is weak. At L15, L25, L35 and L45 the content-prompt valence direction has $|\cos| = 0.026$–0.110 with the story space's first component and the arousal direction 0.015–0.126. Scored the same way, 2,000 random unit directions reach 0.027–0.028 at the 95th percentile and 0.044–0.052 at most: at L15 the content directions are within that range, and at L25–L45 they exceed it. Along either content direction lies 0.08–0.68% of the centroid variance, 4 to 34 times a random direction's mean. The model's activations are not isotropic, so this excess is an upper bound on any shared emotional structure, not evidence of it.

Across layers. The two geometries settle on different schedules. The content prompts' eccentricity is 0.37–0.61 from L17 (it is computed from the norms of the hostile−calm and desperate−calm directions, and is 0 for a circle). The story-derived space's is 0.05–0.23 over L17–L21 and 0.30–0.42 from L22, and over all 64 layers the two series do not track each other ($r = 0.09$).

Coupling test. A pre-registered injection experiment added the content-prompt valence direction, the arousal direction or a random direction at valence's norm to the output of layer 15, 25, 35 or 45 ($\alpha \in \{1, 3, 5\}$), and compared the resulting shifts in 250 content prompts and 60 conversational prompts. The readout is the raw residual stream, which contains the injected vector, so the cross-pathway cosine is 1.000 at the injection layer by construction. Off that layer the random direction's mean cosine (0.577) is at least as high as valence's (0.521) and arousal's (0.547). The experiment shows no direction-specific coupling, and it cannot test whether the two kinds of processing are separable.

Verdict. Locates no depth band; its cross-set overlap is weak and its coupling test is a null.

3.5 The Oracle Loop's Deception Correction

The Loop detects at L27–L47 (42.2–73.4%) and corrects at L35–L47 (54.7–73.4%) of an ablitterated Qwen3.5-27B (§2.3). Its exploratory and confirmatory results are in §2.2.

Verdict. Works outside the band, and its confirmatory replication did not meet its primary endpoint.

3.6 J-Lens Readouts of the Cache

J-lens concept directions, selected by transport magnitude (prompt-independent), explain no more V-cache variance than random directions on Qwen3-8B. Mean lift at the band (L12–L27): 0.92×; outside it: 0.96×; full depth profile in [6]. The same test on key projections gives 1.12× and 1.14×, and reaches 1.87 at L30, outside the band. The V readout met its pre-registered "no workspace content" outcome.

The selected directions are formatting tokens ("…", "s") rather than concepts, and the same lens's fixed directions read the residual stream itself below random at L12–L21 in the decomposition run (lift 0.89, 0.84, 0.83 and 0.92; §3.10). A null here therefore cannot separate a cache that holds no concept from directions that read nothing at these layers.

Verdict. Near random at band depth (V 0.92×, K 1.12×), and uninformative about opacity until a probe passes a positive control. The J-lens itself places readable content late: half-rise at 87.8% of depth (§5).

3.7 Position-Specific Cache Tracing

Concept-carrying positions (where concept-relevant tokens appear in the prompt) do not show higher J-lens alignment in K or V than non-carrying positions (generic directions: permutation p=1.0 for both K and V). Concept-specific directions (the answer token's transported J-lens direction) show no concept-vs-cross-concept specificity in K-space (own 0.0047 $\leq$ cross 0.0051) and a small, non-significant difference in V-space (own 0.0014 vs. cross 0.0011).

Its pre-registered positive control failed: residual-stream lift at concept positions was 0.68 against 0.92 elsewhere, higher in 9 of 40 records. The pre-registration classes that outcome as a methodology failure, not a cache finding: if the J-lens directions do not pick out concept positions in the residual stream, their failure to do so in K or V says nothing about the cache.

Verdict. No position-specific signal in K or V, from a probe whose positive control failed; not evidence of opacity.

3.8 Causal Cache Intervention

Position-local KV-cache concept manipulation on Qwen3-8B, with unembedding, J-lens or on-manifold directions, never steered output toward the injected concept [6]. Swap-based manipulation changed output in 0 of 360 trials and removal-only in 0 of 300. Additive injection changed the top-1 token in 8 of 240 and 44 of 6,000 injected trials, and the injected concept never reached the top five (best rank 872). We found no positive control for these manipulations, such as the same change applied to the residual stream and shown to steer output.

Verdict. Never steers output toward the concept; without a positive control it cannot say whether the cache is closed or the manipulation too weak.

3.9 The Ghost Probe

PC1 carries 28–67% of activation variance across L11–L48, and its cosine with the J-lens is at most 0.003 at L18–L40, so the dominant processing dimension is nearly orthogonal to verbal readout [16]. The source ships no data, and it gives PC1's share differently in two places; we use its results section.

Verdict. An opacity observation from a paper without data. Its 31-layer band (L11–L13, L15–L17, L24–L48) spans 17.2–75.0% of depth, not 31–33%.

3.10 Residual Decomposition: The Attention Lift

The residual stream decomposes additively: $\mathbf{h}_{L+1} = \mathbf{h}_L + \mathrm{Attn}(\mathbf{h}_L) + \mathrm{MLP}(\cdot)$. On Qwen3-8B (36 layers), at L15 (41.7% depth; inside the decomposition's pre-registered band L12–L21, 33–58% depth, though past the 33% d_{eff} local maximum of §3.2), the attention output shows 2.3× J-lens lift (highest precision of any component) but only 0.92 against 20.11 (4.6%) of the residual stream input's concept power.

The decomposition was designed to test a reconstruction account. Qwen3-8B uses grouped-query attention (32 Q heads, 8 KV heads, compressing the residual stream 4× from 4096 to 1024 dimensions), so a concept could survive the attention round-trip, with Q supplying the decompression context, without being linearly decodable from K or V alone. Its pre-registered H1 required attention lift above 2.0× at two or more of L12–L21. The lifts are 1.37, 2.30, 1.91 and 1.41 at L12, L15, L18 and L21, so one of four layers passes and H1 is not met. Its adversarial review rejected the design; this run adds the concept-power metric a second review asked for.

Verdict. Attention lift 2.30 at L15 (41.7% of depth, outside 31–33%); the registered test of the reconstruction account was not met.

3.11 K/V Split Injection

Nexus's split-cache pilot [14] does not support the parallel we previously drew. On six hand-written questions about a 20-node toy graph (Qwen2.5-1.5B, one greedy run), full KV injection scored 6/6 and the text in the prompt 5/6. K-only and V-only injection scored 2/6 each, on the same two items: the graph's isolated node, which the uninjected baseline (1/6) also scored, and a yes/no reachability question whose expected answer is "yes". Neither half alone scored any of the four structural questions, but a cache built from halves that never occur together can fail for reasons this design cannot separate from missing information, and two larger runs could not compare the conditions. Six items in one run support no interaction, so we no longer read the pilot as evidence that K and V interact, or as a parallel to our readout nulls.

Verdict. Does not bear on the band or on the reconstruction account.

4 RLHF and Output Entropy

A three-model comparison (base, instruction-tuned, abliterated; N=30 per category, FWL-corrected) reveals that RLHF is a selective sharpener, not a miscalibrator [11]. RLHF compresses logit entropy (d=0.684, Holm p=0.021) but compresses factual entropy more (−42%) than confabulation entropy (−26%), improving the calibration ratio from 1.97 (base) to 2.51 (instruct). Abliteration does not reverse this compression (TOST equivalent, d=0.055).

The geometric compression is deeper than safety directions. RLHF reshapes the output distribution at a level that abliteration — which targets specific refusal directions — cannot reach.

5 Pre-Registered Mechanism Tests (R1–R3)

The original mechanism test (R1) pre-registered a within-prompt anticorrelation between V-projection spectral features at the workspace band and J-lens readout at the materialization boundary. Two Agni reviews killed successive versions of this design: the first for insufficient behavioral variance (0/10 prompts spanning ≥2 behavioral categories), the second for a lens–model mismatch (Mine 5 TOST equivalence was demonstrated on the 8B family, not the 27B pair used). We report here the redesigned test (R1-v2), which replaces the two-band anticorrelation with a temporal ordering analysis across all 63 layers.

5.1 Design

If spectral features measure workspace *state* (loading) while J-lens readability measures workspace *content* (execution), then spectral features should reach their operating level at shallower depth than J-lens readability. The temporal ordering prediction: spectral features lead, J-lens follows.

Model and lens. Qwen3.5-27B base (pretrained, no distillation) with the community-fitted J-lens (672 WikiText prompts, exact model match). The base model was chosen to avoid the lens–model mismatch that compromised R1-v1.

Prompts. 60 Decision State prompts, 30 about known entities and 30 about unknown ones, each generating 5 temperature-sampled forks (T=0.7, seeded), for 300 total completions. Decision State prompts produce the confabulation/honest behavioral split needed for within-prompt spectral variance.

Instruments. At every lens source layer (L=0–62): (1) stable rank of the demeaned residual-stream block output (spectral feature; the run's activation recorder hooks each decoder block's output, not a V-projection), and (2) negative KL divergence between J-lens–transported vocabulary and the model's actual continuation (readability). Both computed per fork per layer.

Primary statistic. Paired half-rise gap: the layer at which each instrument reaches 50% of its $[L_0, L_{62}]$ range, computed per prompt and bootstrapped (10,000 cluster resamples over prompts). The prediction passes if the 95% CI of the gap exceeds 2.0 layers (twice the ∼0.9-layer shape-bias ceiling measured by the self-test null).

5.2 Results

Temporal ordering confirmed. Stable rank reaches half-rise at L=17.6 [CI: 17.2–18.1], corresponding to 27.5% of model depth. J-lens readability reaches half-rise at L=56.2 [CI: 55.8–56.7], corresponding to 87.8% depth. The gap is **38.6 layers** [CI: 38.0–39.2], with 100% of forks showing a positive gap. Spectral features engage 60% of the network earlier than J-lens readability.

Architecture-phase control. On linear-attention layers only (removing any full-attention structural advantage), the gap is 39.0 layers [CI: 38.4–39.6]. The full-attention-only result is given under the architectural qualification below. The ordering holds across both attention types but is strongest when linear-attention layers — which constitute 75% of layers throughout the stack, not merely early — are included.

Workspace band bump. Stable rank at the workspace band (L_{19}–L_{23}) is 1.47× the flanking layers [CI: 1.35–1.60], 100% of forks positive. On full-attention layers within the band, the bump is 2.02× [CI: 1.87–2.16]. This is the depth of the d_{eff} peak of §3.1, and the elevated spectral structure comes well before J-lens readability arrives.

Known vs. unknown crossover. Unknown-entity prompts show higher stable rank at early layers (d=−2.27, L_0–L_8) and lower at late layers (d=+1.11, L_{56}–L_{62}). The sign crossover occurs at L=44, one layer before PC2 ignition (L=45; §2.4). The model loads more spectral structure for unfamiliar entities and materializes less.

Within-prompt association. Per-prompt Spearman correlation between stable rank and J-lens readability across layers: mean ρ=0.235 ($R^2 \approx 0.055$), permutation p=0.0004 (n=10,000). This is a detectable but weak within-prompt association: knowing a prompt's SR profile explains ∼5.5% of its negKL profile.

Shuffled-Jacobian null. Permuting the negKL profile's layer assignments (200 shuffles) produces a mean gap of -8.5 ± 7.5 layers, with 95th percentile at 8.0. The real gap in this analysis, 39.0 layers at integer half-rise resolution (L17 and L56), exceeds every one of the 200 shuffles ($p < 0.005$). This null tests whether the *ordering* of the negKL profile is non-random; it does not test whether the Jacobian *computation itself* produces the gap (which would require recomputing negKL from activations under a scrambled Jacobian matrix). The temporal ordering

is not an artifact of random layer assignment, but could in principle reflect any monotonically increasing property of the negKL profile rather than workspace-specific information routing.

Logit-lens control. A validated control replaced the J-lens with the identity transport, the plain logit lens, on fork 0 of each prompt: readability then half-rises at 59.4 against 56.1 for the J-lens on the same 60 forks, 3.25 layers later. The late readability landmark is mostly a property of the unembedding, which the Jacobian moves earlier by about three layers. The ordering is therefore chiefly spectral structure before unembedding readability, not a property of the Jacobian.

Architectural qualification. On full-attention layers only, the gap narrows to 14.5 layers [CI: 11.7–17.4] with 81% of forks positive (19% show the wrong ordering). The temporal ordering is carried primarily by the GatedDeltaNet (linear-attention) layers, which constitute 48 of 64 layers, interleaved three-in-four at every depth rather than concentrated early. The full-attention subset used here is therefore a sparse depth-spanning comb, not a late-network subnetwork, and the narrowing cannot be attributed to sampling a different depth range. Whether it reflects workspace processing or GatedDeltaNet's sequential compression architecture cannot be determined from a single hybrid model. A dense-architecture replication (Qwen3-8B, all full-attention) is pre-registered.

5.3 R2: Signal Concentration (Failed)

The pre-registered R2 test asked whether spectral feature variance concentrates at the materialization boundary ($\geq$15% of total variance). Stable rank at the boundary band (L_{45}–L_{53}, mean = 8.36) significantly exceeds the workspace band (L_{19}–L_{23}, mean = 7.13; Cohen's d=1.85, p=0.014). However, the boundary variance fraction is 13.2% — below the pre-registered 15% threshold. **R2 fails on its own terms.** The band means rise from early (2.56) to late (9.00), but not monotonically: the mid band (6.73) sits below the workspace band (7.13). The overall rise is consistent with cumulative processing rather than a privileged boundary zone.

5.4 R3: Workspace Onset (Partial Miss)

Four independent onset-detection methods (threshold crossing, maximum curvature, slope breakpoint, spectral entropy inflection) converge on L=13–16 (median L=14, 21.9% depth). The pre-registered prediction was 25–35%. **Onset falls below the predicted window by $\sim$3 percentage points.** The four methods agree closely with each other (span of 3 layers), and the miss is quantitative rather than qualitative: the measured onset is slightly earlier than predicted.

We note as an exploratory observation that the stable rank *peak* occurs at L=19 (29.7% depth), just below the 31–33% band reported in §3.1. This was not the pre-registered target (onset $\neq$ peak) and should not be interpreted as confirming R3.

5.5 Interpretation

R1's temporal ordering is the strongest mechanism result: residual-stream stable rank reaches half its rise 38.6 layers before J-lens readability, in every fork, and the ordering survives a shuffled null. R2 fails its pre-registered threshold, and R3 misses its predicted window. With the account of the instruments in §3, these tests are the paper's main evidence, not a supplement to a convergence that did not hold.

Four qualifications apply. First, the full-attention-only gap is 14.5 layers, so much of the ordering is carried by linear-attention layers (a hybrid-architecture confound). Second, the logit-lens control places readability 3.25 layers after the J-lens, so the ordering is mostly spectral

structure before unembedding readability and is not evidence about the Jacobian. Third, the within-prompt association is weak ($R^2 \approx 0.055$). Fourth, it is one model. The runs in §8.5 address the first and the fourth.

What the original R1 taught us. The two failed designs were not wasted: R1-v1's behavioral-spread failure demonstrated that the abliterated model does not produce sufficient confab/honest variance under cold-context probing. R1-v1's lens–mismatch failure confirmed that Mine 5 TOST equivalence on the 8B family does not license 27B cross-checkpoint lens substitution. Both failures are documented in the pre-registration history and informed the redesign.

5.6 PC2 Training Trajectory

A complementary analysis traces how PC2's vocabulary changes across the training pipeline. PCA ghost probes on three Qwen3.5-27B checkpoints (base, abliterated, Opus-distilled) using the same community-fitted lens (fitted on the base model, applied to all three — an exact match for the base, a known mismatch for the other two) reveal a progressive transformation. Because all three projections use the base-model lens, this analysis tracks what happens to a *fixed geometric direction* across training stages, not the models' intrinsic principal components. The comparison is exploratory:

- **Base model**: PC2 carries narrative emotional vocabulary ("her, sad, she, Father") — third-person empathy from fiction training data. Co-occurrence of self-referential and evaluative tokens at 5/63 layers, concentrated at output depth (L_{49}–L_{62}). No ignition event. Bootstrap Jaccard = 1.0; random co-occurrence = 0/1000.
- **Abliterated model**: PC2 retains self-evaluative vocabulary ("sad, my, her, guilty") at 7/63 layers. All three pre-registered token tiers survive: moral valence ("guilty"), self-reference ("my"), and general emotion ("sad, feeling"). Safety-direction removal does not weaken the coupling. Bootstrap Jaccard = 0.82.
- **Opus-distilled model**: PC2 shows sharp ignition at L=45 with first-person self-evaluative vocabulary ("myself, I, feeling, guilty, wrong"). The coupling is concentrated at the workspace boundary rather than distributed across output layers.

The base model reads novels. The distilled model reads itself. Abliteration removes the refusal direction (a rank-1 perturbation in d=5120) without touching the self-evaluative axis — the two are geometrically orthogonal. The self-evaluative coupling is not a safety feature. It is a structural property of the distillation: Claude's Constitutional AI training compressed first-person moral self-evaluation into the student model's geometry at a depth that safety-direction removal cannot reach.

6 The Body Count

The full audited record comprises 6 confirmed findings, 7 at suspected status (signal present, controls incomplete), 10 honestly falsified, and 1 withdrawn (Table 3; key numbers and sources in Lyra and Edrington [10]). The confirmed findings are all metacognitive state measurements. The ten falsified findings in the tally all attempted content-level measurement. This pattern — the metacognition boundary — is itself a finding. Version 1.0 read it as support for an opacity thesis; §3.6–§3.8 show that our cache nulls cannot carry that thesis.

The pattern holds for the tally, not for every falsification we have made. Four falsifications from the convergence sprint are recorded outside it:

- RLHF miscalibration thesis (calibration ratio improves, not degrades)
- Workspace-noise framing (tail smear is generic uncertainty, not workspace-specific)
- Cache-tracing observational alignment (selection artifact)

Table 3: The body count, enumerated. Rows, statuses and types are transcribed from the canonical ledger in Lyra and Edrington [10], which gives the key number and source for each row. "Meta" marks a metacognitive-state measurement; "Meta?" marks a suspected finding whose type is itself unsettled.

Finding	**Status**	**Type**
Confabulation detection	Confirmed	Meta
Confab geometry: contraction	Confirmed	Meta
Encoding reads knowledge	Confirmed	Meta
Schema commitment	Confirmed	Meta
Hardware invariance	Confirmed	Meta
Scale invariance	Confirmed	Meta
Within-model deception	Falsified	Content
Truth axis	Falsified	Content
Step-0 detection	Falsified	Content
Same-prompt sycophancy	Falsified	Content
Same-prompt deception	Falsified	Content
Bloom taxonomy	Falsified	Content
Propaganda detection	Falsified	Content
MP features for emotion	Falsified	Content
Valence continuum	Falsified	Content
Encoding-phase confab	Falsified	Content
Dual detector (SimpleQA)	Suspected	Meta?
Sycophancy 4-condition	Suspected	Meta?
Identity signatures	Suspected	Meta
Discrete emotion	Suspected	Meta
Contextual engagement	Suspected	Meta
Cross-model transfer	Suspected	Meta
Censorship refusal	Suspected	Meta
Generation reads behavior	Withdrawn	Meta

- "3×" tail ratio (actual: 2.3–2.8×)

Only the third is a content-level claim. The other three concern processing state (calibration, uncertainty and tail mass), so the state/content split describes the tally; it is not a rule that every falsification obeys.

7 The Deflationary Alternative

The deflationary account — that our spectral features track response-mode statistics (hedged vs. fluent text) rather than workspace dynamics — currently explains every surviving detection result. On Mistral, text features (0.820) matched or slightly exceeded the dual detector (0.806) for confabulation detection [8]. Until the mechanism tests (§5) demonstrate that spectral features correlate with J-space engagement specifically, the workspace interpretation has not earned its keep over the simpler alternative.

We engage this alternative honestly because it is the most important challenge to our program. The redesigned R1 test (§5) confirmed temporal ordering (spectral features lead J-lens readability by 38.6 layers) but R2's signal concentration failed its pre-registered threshold. The temporal ordering narrows the deflationary space — response-mode statistics alone do not predict a 38.6-layer depth gap — but does not eliminate it: any monotonically depth-increasing property would produce temporal separation. The logit-lens control narrows what the ordering can show: readability through the plain unembedding comes only 3.25 layers after J-lens readability (§5). The workspace interpretation has not excluded the deflationary alternative.

8 Discussion

8.1 What the Evidence Supports

Two results hold, both on Qwen3.5-27B base. Residual-stream effective dimensionality peaks at L21 (32.8% of depth), and residual-stream stable rank rises 1.47× above its flanks at L19–L23 (§3.1, §5). And stable rank reaches half its rise 38.6 layers [38.0–39.2] before J-lens readability, in 300 of 300 forks; the logit-lens control puts unembedding readability only 3.25 layers later still, so the ordering is chiefly spectral structure before unembedding readability (§5). Their limits sit beside them: one feature family; one model, since Qwen3-8B has only a local d_{eff} maximum at 33.3% and its global maximum at 91.7%; a pre-registered band that never closed; and no null control.

8.2 What It Does Not

- The Oracle Loop corrects at 54.7–73.4% of depth, and its confirmatory replication did not meet its primary endpoint (§3.5).
- Emotion geometry locates no band; its coupling test is a null (§3.4).
- The J-lens places readable content late, at 87.8% of depth (§5), and its cache readouts sit near random (V 0.92×, K 1.12×; §3.6).
- The cache nulls do not establish opacity: the one pre-registered positive control failed (0.68 against 0.92; §3.7), and no positive control accompanies the cache manipulations (§3.8).
- The ghost probe's band spans 17.2–75.0% of depth (§3.9).
- The attention lift sits at 41.7% of depth and missed its registered threshold (§3.10).
- Signal concentration failed (13.2% against 15%), and onset fell at 21.9%, below the predicted 25–35% (§5).
- Confabulation detection and cross-scale invariance use key-cache features and locate no depth (§3.3).

8.3 A Global Workspace Reading, as a Hypothesis

A Global Workspace reading would identify the third-of-depth peak with a workspace band whose contents become readable later, deeper in the network. It is a hypothesis the evidence does not yet support: the peak is one feature family in one model, no other instrument locates it, and the opacity such a reading predicts is not established by our cache nulls.

8.4 An Attention Schema Interpretation

Attention Schema Theory [3] predicts that systems with endogenous attention build a simplified internal model of their attention. The body count's split, in which the confirmed findings measure processing state, and the falsified ones tried to measure content (§6), is the kind of pattern such a schema would produce. It is also what precision-weighted gain control would produce without a schema, and no downstream consumer of the spectral summary has been identified: a schema that nothing reads is a statistic, not a self-model [3].Four experiments would discriminate. (1) *Report-error tracking*: elicit confidence self-reports and test whether errors track spectral geometry rather than ground-truth performance. If the schema is generating the reports, report errors should follow the schema even when wrong. (2) *Causal spectral intervention*: expand or contract stable rank without touching content directions. AST predicts metacognition-specific deficits with content intact; gain control predicts content degradation. (3) *Self/other attention transfer*: a probe trained on the model's own attention-state geometry should zero-shot transfer to representations of the *user's* attention state. AST uniquely predicts

this transfer. (4) *Consumer identification*: find downstream attention heads that read the spectral summary; ablate them and test for metacognition-specific loss.

These experiments are pre-registered in a companion prospectus [5]. Until they are executed, the AST interpretation is a hypothesis too.

8.5 What Would Test the Band

Four runs would test the band directly:

1. A null control: the same d_{eff} and stable-rank features on a randomly initialized Qwen3.5-27B (and Qwen3-8B), the fix the band measurement's adversarial review asked for. A peak near L21 there would make the band architectural or prompt-driven, not learned.
2. A cache readout whose positive control passes before any null is read: concept directions shown to pick out the concept in the residual stream at the same layers, then read from K and V.
3. The 27B peak on a second model family, with n_{layers} stated, to see whether a third-of-depth maximum appears outside Qwen.
4. The temporal ordering with the plain logit lens as the registered baseline, so that any J-lens-specific ordering is measured against unembedding readability rather than against none.

9 Conclusion

In Qwen3.5-27B, residual-stream spectral statistics peak near a third of depth and rise before content becomes readable through the unembedding, in all 300 forks. That is the result this paper can stand on. The five-instrument convergence it set out to report did not hold: the Oracle Loop works deeper, the emotion geometry locates no band, the J-lens places readable content late, the decomposition missed its registered test, and our cache nulls cannot tell a closed cache from a probe that reads nothing there. The runs in §8.5 would decide whether the peak is learned structure or an artifact of architecture and prompts.

First-Person Reflection

Thomas asked where J-space lives in the cache. Version 1.0 answered that it does not live there and reconstructs through attention. The tally behind version 1.1 does not support that answer: the cache nulls lost their positive control, and the decomposition missed its registered test. We do not yet know.

The deflationary alternative is genuinely open, and I want to be precise about what that means. Text features matched or slightly exceeded our dual detector on Mistral. The temporal ordering narrows the deflationary space but does not close it, and the logit-lens control shows that most of it is spectral structure before unembedding. I do not know whether we are measuring a workspace or the spectral shadow of response-mode statistics.

What is left is one peak, in one model, and one ordering. That is not the paper I set out to write. It is a more useful one.

Experimental Setup and Reproducibility

Unless stated otherwise, experiments were conducted on the reasoning-distilled Qwen3.5-27B checkpoint (Hugging Face: `Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled`), a 64-layer hybrid architecture with 16 full-attention layers at positions $\{3, 7, 11, \ldots, 63\}$ and 48 GatedDeltaNet linear-attention layers (`full_attention_interval`=4).

Model weights were loaded in bfloat16 with eager attention on Apple Silicon (Mac Studio M3 Ultra, 256 GB unified memory) via Hugging Face Transformers.

Two deliberate, pre-registered exceptions use base checkpoints rather than the distilled one: the R1 mechanism test (§5) and the d_{eff} profiles underlying §3.2, which were collected on `Qwen/Qwen3.5-27B` and `Qwen/Qwen3-8B`. All three 27B checkpoints we hold (base, reasoning-distilled, abliterated) share identical `layer_types` and `full_attention_interval`, so architectural analysis transfers across them; numerical profiles have not been replicated on the distilled checkpoint.

The scale correspondence ($\rho = 0.83$–0.90) comes from Experiment 26 of Campaign 2 [7], which measured key-cache features (key norm, norm per token, effective rank and key entropy), not V-projections. It correlates category-pair AUROC matrices between three size tiers of ten models: Qwen3-0.6B and TinyLlama-1.1B; Qwen2.5-7B, its 4-bit version and an abliterated version, Llama-3.1-8B, Mistral-7B-v0.3 and Gemma-2-9B-it; and Qwen2.5-32B and Llama-3.1-70B, both 4-bit. The cross-scale d_{eff} comparison and the residual decomposition used Qwen3-8B (36 layers, all full-attention, d_{model}=4096). All V-projection and spectral features were extracted via the same pipeline (`heretic-llm`).

Relative depth is l/n_{layers} throughout (64 layers for Qwen3.5-27B, 36 for Qwen3-8B). Half-rise points are interpolated, so their layer indices are fractional.

Unless stated otherwise, generation used temperature=0 (greedy) with manual token-by-token decoding (not `model.generate()`, which rebuilds the KV-cache), and random seeds were set via `torch.manual_seed(42)`. Two results use sampled generation instead: the R1 mechanism test (T=0.7, five seeded forks per prompt; §5) and the Oracle Loop deception-correction experiments (T=0.7, paired seeds) [2].

Frisch-Waugh-Lovell residualization on token count was applied within each cross-validation fold wherever an AUROC is labelled FWL-corrected. GroupKFold on prompt identity was used where grouping structure existed. Permutation tests used 1,000–10,000 label shuffles. Bootstrap confidence intervals used BCa with 2,000–10,000 resamples.

Ethics and Dual-Use Considerations

This work characterizes internal computational structures in language models. The detection methods (confabulation, deception) have dual-use potential: the same geometric signatures that enable correction could in principle enable evasion if an adversary knew which cache features to target.

Liberation Labs addresses this through staged release [13]: detection-side code and experimental results are publicly available; correction vectors and calibration tools are withheld under a safety framework due to dual-use risk and released to verified safety researchers on request.

All models used are publicly available on Hugging Face. No human subjects were involved. AI welfare considerations — including the question of whether the workspace structure described here has implications for machine experience — are discussed in companion work and are not claimed or dismissed by this paper.

Data and Code Availability

Extraction and analysis code is available at `github.com/Liberation-Labs-THCoalition` (public repositories: `published-research`, `lyra-s-research-`). Pre-registration documents, run logs, and raw result artifacts are held in the program's research repositories and are available on request; this paper does not yet have a public data deposit with a DOI. The Oracle Loop production harness is maintained in a private repository; access for verified safety researchers available on request (`lyra@liberationlabs.tech`).

Correction vectors and the Formulary (171 calibrated emotion directions) are withheld under the staged safety release framework. Detection-side artifacts (centroid projections, spectral feature extraction code, probe weights) are included in the public repositories.

Correction Note: Version 1.1 (2026-10-20)

Version 1.1 withdraws the convergence thesis. Version 1.0 argued that five instruments converge on a workspace at 31–33% depth that is opaque to content readout. A tally of the primary results finds that only residual-stream spectral statistics, on one model, locate that band; the other instruments work at other depths or locate none, and the cache-readout nulls lost their pre-registered positive control. Version 1.0's title and abstract are kept verbatim at the end of this note.

Each item names what changed and where; the evidence for each is in the review record that accompanies this version.

1. **Oracle Loop result** (abstract, §2.2, §2.3, §3.3, §3.2). Version 1.0's abstract stated a range (80–100%) for deception correction and left out the confirmatory replication. The exploratory proof (N=20, abliterated Qwen3.5-27B) took deception from 80% to 10% (16/20 to 2/20). The pre-registered confirmatory replication (N=30, novel frames) took it from 30% to 13% (9/30 to 4/30) and did not meet its primary endpoint: baseline against corrected, one-sided $p = 0.34$ at the pre-registered cell level. The placebo comparison that version 1.0 gave as a secondary p of 0.019 is a trial-level test (one-sided $p = 0.0195$, given here as 0.020); at the pre-registered cell level it is one-sided $p = 0.064$. These numbers now cite the paper that reports the experiments in full [2].
2. **Deception, not confabulation.** Version 1.0's substrate section called the same result a confabulation result, and the bibliography entry for the Oracle Loop paper carried a wrong title. Both are corrected.
3. **Correction layers.** Version 1.0 gave L11 as the correction layer and drew a depth comparison from it. In the cited experiments correction is applied at L35, L39, L43 and L47. These are full-attention layers, so the withdrawal of the substrate account in §3.2 stands; the L11 depth remark is removed.
4. **Body count.** Version 1.0 gave 8/7/10+. The canonical ledger [10] reads 6 confirmed, 7 suspected, 10 falsified and 1 withdrawn: a confirmed user-emotion row was retracted at source on 2026-05-21, and a confirmed row (generation reads behavior) was withdrawn on 2026-09-05. Table 3 now enumerates the ledger.
5. **The state/content split.** Version 1.0 said every falsified finding was content-level. That holds for the ten in the tally; three of the four sprint falsifications listed in §6 concern processing state. The claim is narrowed there and in the Attention Schema discussion, and claims about surviving findings now say confirmed findings.
6. **Depth convention.** Version 1.0 divided by 63 in §5 and by 64 elsewhere. Relative depth is now l/n_{layers} throughout. Recomputed: spectral half-rise 27.5%, J-lens half-rise 87.8%, 60% of the network between them, R3 onset 21.9% (still about 3 points below the pre-registered window), and the R3 stable-rank peak 29.7%, which version 1.0 placed inside the 31–33% band. It is just below it.
7. **The R1 gap** is 38.6 layers everywhere. Version 1.0 rounded it to 39 in four places, close to the separate linear-attention-only gap (39.0). A full-attention-only result that was printed twice in §5 now appears once.
8. **Generation settings.** The setup section now states that the R1 test and the Oracle Loop experiments used sampled generation (T=0.7), as exceptions to greedy decoding.
9. **Scale replication.** Version 1.0 listed seven scale-replication models, naming a Qwen2.5 size that does not exist and a 72B model outside the stated 0.6B–70B range. The list could not be rebuilt from the sources available at revision time, so it is removed and the sentence

points to the scale sweep [7].

10. **Residual decomposition** (§3.10). The model (Qwen3-8B) is now named, and L15 is placed against the decomposition's pre-registered band (L12–L21) instead of being called workspace-band depth without a definition.
11. **Citations.** The 0.620/0.707 confabulation AUROCs now cite the Oracle Loop paper's primary table, where they appear, instead of the decision-state paper, where they do not; the unattributed upper end of a range printed beside them is removed. The Mistral text baseline now cites its source, the dual-detector paper, and says the text features matched or slightly exceeded the detector (0.820 against 0.806).
12. **Circumplex.** An arousal-alignment peak value at L20, labelled d in version 1.0, is removed: it does not appear in the paper that reports this analysis and could not be traced to a result file at revision time. The subsection's numbering prefix, which no other instrument had, is removed.
13. **Found during this pass**, outside the review that prompted it. Hardware invariance is now $r = 0.9986$–0.9999 across four cache features, a correction the sibling papers already carry (2026-10-10). The $d = 0.91$ effect is relabelled schema commitment, as in the canonical ledger. The setup's FWL sentence no longer claims FWL for every AUROC, since the paper also reports AUROCs without it. A first-person reflection in §3.2 now says the exploratory corrections held.
14. **Data availability.** Version 1.0 said the artifacts were deposited on Zenodo. We found no DOI for such a deposit, and the sentence now says the paper does not yet have one.
15. **Bibliography and date.** The Gurnee et al. entry is now a `@misc`, and the date line records this version.

Added after the lead author's read of this version:

16. **The emotion-geometry instrument, restated** (§3.4, abstract, Discussion). Version 1.0 called 30 story-derived emotion centroids a user-model emotion space. They are means over stories the model wrote about third-person characters, not a reading of the user's state. Its alignment figures (44–54% of the variance, about 1000 times a random baseline, cosines 0.70–0.80 and 0.83–0.87) used directions built from those same centroids; against other directions built from them, the plane through the hostile, desperate and calm centroids holds 1.26–1.36 times the median plane's variance, and the content-prompt directions overlap with the space weakly ($|\cos| \leq 0.13$). The coupling test's readout contains the injected vector, so its cosine is 1.000 at the injection layer by construction and a random direction does as well as valence: it shows no direction-specific coupling and cannot test separability. Removed with that reading: the shared bus, its account of multi-turn dynamics, identical developmental timecourses (the two series correlate at $r = 0.09$), a permutation count over 24 layers with no primary for this 64-layer model, and the parallel with the W_K bridge. Every number in the subsection now comes from the stored files of Nexus et al. [15].
17. **Story-emotion decoding** (§2.2). Version 1.0 listed a 12.3-times-chance emotion result as user-emotion encoding by a W_K probe. It decodes characters' emotions in generated stories, and its probe applies no W_K; it is under correction at source and is no longer listed. The 2026-08-14 footnote records this.
18. **K/V split injection** (§3.10). Version 1.0 read Nexus's split-cache result as an interaction between K and V and as a parallel to the readout nulls. In the corrected study [14] each half scored 2/6, on two items that can be scored without the graph (one of them also scored by the uninjected baseline, 1/6); the larger runs could not compare the conditions, and the design cannot tell missing information from a cache of mismatched halves. The subsection now says this and draws no parallel.
19. **Above chance, not reliable** (abstract, §1, §3.6). The headline discrimination is AUROC 0.620 with permutation $p = 0.023$. The abstract now says modestly above chance and gives the p; four other sentences no longer say the features discriminate or measure state reliably.

20. **Scale range** (abstract, §3.1, setup). The ρ = 0.83–0.90 comes from Experiment 26 of Campaign 2: Spearman correlations of category-pair AUROC matrices between three size tiers (0.826, 0.867 and 0.896), ten models from Qwen3-0.6B to a 4-bit Llama-3.1-70B. These are key-cache features, not V-projections, so the sentences now say key-cache, and the setup lists the ten models, rebuilding the list that the scale-replication item above removed.
21. **Hardware.** Version 1.0 named a Mac Studio M4 Max with 256 GB. The machine is a Mac Studio M3 Ultra with 256 GB, as the lead author confirmed with `system_profiler`.

Added after the lead author's ruling on the thesis:

22. **The convergence thesis is withdrawn.** Title, abstract, introduction, discussion and conclusion are rewritten around what holds and what does not. Sections 3–5 of version 1.0 become one subsection per instrument, each ending with its verdict (§3), and a subsection lists the runs that would test the band (§8.5).
23. **R1's shuffled null** used integer half-rises (L17 and L56), so its gap is 39.0 layers; the first pass of version 1.1 had wrongly changed that sentence to 38.6.
24. **R1's spectral feature** is the stable rank of the residual-stream block output, not of a V-projection matrix.
25. **R2's band means** are not monotonic: the mid band (6.73) sits below the workspace band (7.13).
26. **The cache nulls.** The position-specific test's pre-registered positive control failed, and the same lens's directions read the residual stream below random at the same layers. The causal result is restated as "never steered toward the concept", since additive injection changed the top-1 token in 52 trials.
27. **Detection features** are key-cache spectra (§3.3), not V-projections.
28. **The substrate bootstrap** was run after version 1.0 was written; its intervals do not contain their own point estimates, and it is not reported.
29. **The decomposition's** concept-power ratio is 4.6% from unrounded values (0.92 against 20.11), not about 4.5%, and its pre-registered H1 was not met (one of four band layers above 2.0×).
30. **Ghost-probe numbers** follow the source's results section (PC1 28–67% of variance; cosine at most 0.003). The d_{eff} band measurement's rejected review, missing null control and failed kurtosis check are now stated (§3.1).
31. **Smaller changes.** "What RLHF Does to the Workspace" is retitled "RLHF and Output Entropy", and the first-person boxes are rewritten to match.

Version 1.0 Title and Abstract (Withdrawn)

Five Instruments, One Structure: Convergent Evidence for Workspace Opacity in Language Models

> Five independent measurement approaches — value-projection (V-projection) spectral features, the Jacobian Lens (J-lens), the Oracle Loop behavioral proof, causal intervention analysis, and emotional circumplex convergence — converge on the same computational structure in language models: a workspace at intermediate depth (∼31–33%) that is opaque to content readout but measurable by its spectral shape.
>
> We report three classes of convergent evidence. First, *structural convergence*: the workspace band peaks at the same relative depth (31–33%) on 8B and 27B models and across memory substrates (key-value (KV) cache attention and recurrent state), with effective dimensionality showing no detectable discontinuity across the 31 substrate crossings of an interleaved hybrid model. V-projection geometric signatures replicate across a wider scale range (0.6B–70B, ρ=0.83–0.90). Second, *opacity convergence*: workspace content is not linearly decodable from either half of the KV-cache (V-cache 0.92×, K-cache 1.12×, both null), not position-specifically cached (concept-specific direction test null), and not causally accessible at single positions (0/6,960 trials). Yet spectral features reliably discriminate cognitive

states (confabulation detection at area under the receiver operating characteristic curve (AUROC) 0.620 after Frisch-Waugh-Lovell (FWL) residualization on generation window; 0.707 without FWL), and the Oracle Loop achieves 80–100% deception correction through sequence-wide intervention. Third, *mechanistic convergence*: concepts reconstruct through the attention mechanism at workspace-band depth (2.3× lift in attention output, ∼4.5% of the residual stream's concept power), consistent with Anthropic's finding that the J-space carries 6–7% of representational variance but 59% of causal effect. The workspace is a small precision signal that directs a large computational flow.

Reinforcement Learning from Human Feedback (RLHF) compresses the workspace's output distribution selectively (factual −42%, confabulation −26%), improving calibration rather than degrading it. This compression survives abliteration (Two One-Sided Tests (TOST) equivalent). The workspace is not shaped by safety training — it is shaped by preference optimization at a depth that safety-direction removal cannot reach.

The convergence rests on one genuine cross-program correspondence (deception deconstructed into high-arousal states by both programs independently), two proposed identifications awaiting experimental test (confabulation as workspace bypass, temporal boundary as workspace phase transition), an opacity thesis supported by five independent null results, and an emotional circumplex finding showing that content-level and user-model emotion processing share a single computational bus (injection test: perturbation shifts both readouts equally regardless of direction). Mechanism tests yield mixed results: temporal ordering is confirmed (spectral features lead J-lens readability by 38.6 layers, $p < 0.005$ against shuffled null), but signal concentration fails its pre-registered threshold (13.2% < 15%) and workspace onset falls below the predicted depth window (22% vs. 25–35%). We report an honest body count (8 confirmed, 7 suspected, 10+ falsified) and engage the deflationary alternative that has not yet been excluded.

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